

P1

$$a) \quad \begin{aligned} \dot{\theta}_1 &= \omega + H(\theta_2 - \theta_1) + H(\theta_3 - \theta_1) \\ \dot{\theta}_2 &= \omega + H(\theta_1 - \theta_2) + H(\theta_3 - \theta_2) \\ \dot{\theta}_3 &= \omega + H(\theta_1 - \theta_3) + H(\theta_2 - \theta_3) \end{aligned}$$

H is odd, periodic (similar to sine function)

$$\text{Let } \phi = \theta_1 - \theta_2, \quad \psi = \theta_2 - \theta_3 \quad \Rightarrow \begin{cases} \theta_2 - \theta_1 = -\phi \\ \theta_3 - \theta_1 = -\psi - \phi \\ \theta_1 - \theta_3 = \phi + \psi \end{cases}$$

$$\Rightarrow \begin{cases} \dot{\theta}_1 = \omega + H(-\phi) + H(-\psi - \phi) \\ \dot{\theta}_2 = \omega + H(\phi) + H(-\psi) \\ \dot{\theta}_3 = \omega + H(\phi + \psi) + H(\psi) \end{cases}$$

using H being odd, we have $H(-x) = -H(x)$

$$\Rightarrow \begin{cases} \dot{\theta}_1 = \omega - H(\phi) - H(\psi + \phi) \\ \dot{\theta}_2 = \omega + H(\phi) - H(\psi) \\ \dot{\theta}_3 = \omega + H(\phi + \psi) + H(\psi) \end{cases}$$

For the 2-frog model, $\begin{cases} \dot{\theta}_1 = \omega + H(\theta_2 - \theta_1) \\ \dot{\theta}_2 = \omega + H(\theta_1 - \theta_2) \end{cases}$,

we have similarly

$$\begin{cases} \dot{\theta}_1 = \omega + H(-\phi) = \omega - H(\phi) \\ \dot{\theta}_2 = \omega + H(\phi) = \omega + H(\phi) \end{cases}$$

b) Now, let $H = a \sin \theta$

(i) We compute the phase difference rate of change, that is,

$$\dot{\phi} = \dot{\theta}_1 - \dot{\theta}_2 = -H(\phi) - H(\phi) = -2H(\phi) = -2a \sin \phi$$

We seek equilibria of phase difference ϕ . Since Aihara found in the croaks to alternate, we expect a stable equilibrium at a phase difference of π , corresponding to half a period of H .

→ (i) [cont.]

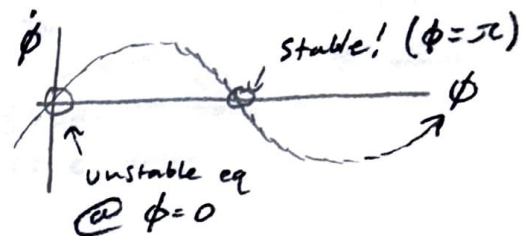
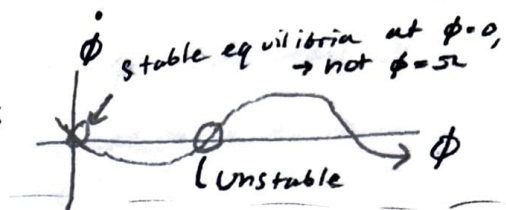
$$\dot{\phi} = -2a \sin \phi \rightarrow \text{equilibria at } \phi \in \{n\pi : n \in \mathbb{N}\}$$

We consider scenarios for $\text{sgn}(a) = \pm 1$:If $a > 0$, $\text{sgn}(a) = +1$, then $\dot{\phi}$ follows

or

If $a < 0$, $\text{sgn}(a) = -1$, then $\dot{\phi}$ follows

⇒ using the slope $\frac{d\dot{\phi}}{d\phi}$ at the equilibria to determine stability, we find that the sign of a must be negative to recover the stable anti-phase crouching solution.



→ (ii) We return to the 3 frog system, using the same technique to reduce the system to a function of the phase differences:

From (1a)
$$\begin{cases} \dot{\theta}_1 = \omega - H(\phi) - H(\psi + \phi) \\ \dot{\theta}_2 = \omega + H(\phi) - H(\psi) \\ \dot{\theta}_3 = \omega + H(\phi + \psi) + H(\psi) \end{cases}$$

$$\dot{\phi} = \dot{\theta}_1 - \dot{\theta}_2 = (\omega - H(\phi) - H(\psi + \phi)) - (\omega + H(\phi) - H(\psi))$$

$$\Rightarrow \dot{\phi} = H(\psi) - H(\psi + \phi) - 2H(\phi)$$

Similarly,

$$\dot{\psi} = \dot{\theta}_2 - \dot{\theta}_3 = H(\phi) - H(\phi + \psi) - 2H(\psi)$$

Notice that $\dot{\phi} = f(\psi, \phi)$ and $\dot{\psi} = f(\phi, \psi)$, i.e., the variables are symmetric in $\dot{\phi}, \dot{\psi}$.

$$\dot{\phi}/a = \sin \psi - \sin(\psi + \phi) - 2\sin \phi = \sin \psi - \sin \psi \cos \phi - \sin \phi \cos \psi - 2\sin \phi$$

$$\dot{\psi}/a = \sin \phi - \sin(\psi + \phi) - 2\sin \psi = \sin \phi - \sin \psi \cos \phi - \sin \phi \cos \psi - 2\sin \psi$$

$$\Rightarrow \dot{\phi}/a = \sin \psi (1 - \cos \phi) - \sin \phi (\cos \psi + 2)$$

$$\dot{\psi}/a = \sin \phi (1 - \cos \psi) - \sin \psi (\cos \phi + 2)$$

$$\Rightarrow \phi^*, \psi^* = n\pi, \text{ or: } \sin \phi (1 - \cos \psi) = \sin \psi (2 + \cos \phi)$$

$$\Rightarrow \frac{1 - \cos \psi}{\sin \psi} = \frac{2 + \cos \phi}{\sin \phi} \Rightarrow \psi^* \pm \arccos \left(\frac{2 + \cos \phi}{\sin \phi} \right)$$

this term is always positive, so need $\sin \phi = \sin \psi = 0$

→ (ii) [cont.]

Thus, the equilibria are

$$\phi^*, \psi^* = n\pi, \text{ or}$$

$$\psi^* = \pm \arccos\left(\frac{-4 \cos \phi - \cos 2\phi - 4}{4 \cos \phi + 5}\right)$$

if $\phi = n\pi$,

$$\psi^* = \pm \arccos\left(\frac{\pm 4 - 5}{\pm 4 + 5}\right)$$

symmetric solution exists for ϕ^* in terms of ψ

~~(*) we also know that if $\phi^* = k\pi$ where k is odd, then $\psi^* = 2\pi$ where k is even (i.e., ϕ^* and ψ^* cannot both be~~

Experimentally, we have equilibria at

$$\psi^* = \phi^* = 2\pi/3. \text{ This is verified below:}$$

$$\phi(\phi^*, \psi^*) = a(4 \sin 2\pi/3 - \sin 4\pi/3 - 2 \sin 2\pi/3) = a(\sqrt{3}/2, \sqrt{3}/2 - \sqrt{3}) = a(0) = 0, \text{ as expected!}$$

... and similarly for ψ .

→ (iii)

We have

$$\dot{\phi} = a(\sin \psi - \sin \psi + \phi - 2 \sin \phi)$$

$$\dot{\psi} = a(\sin \phi - \sin \psi + \psi - 2 \sin \psi)$$

Stability matrix (let $\vec{\theta} = \begin{pmatrix} \phi \\ \psi \end{pmatrix}$) $\rightarrow \dot{\vec{\theta}} = \underline{A} \cdot \vec{\theta}$

$$\underline{A} = \begin{pmatrix} \partial_{\phi} \dot{\phi} & \partial_{\psi} \dot{\phi} \\ \partial_{\phi} \dot{\psi} & \partial_{\psi} \dot{\psi} \end{pmatrix} = a \begin{pmatrix} -\cos(\psi + \phi) - 2 \cos \phi & \cos \psi - \cos(\psi + \phi) \\ \cos \phi - \cos(\psi + \phi) & -2 \cos \psi - \cos(\psi + \phi) \end{pmatrix}$$

There is no restriction on which 2 frogs couple in the pair + 1 antiphase solution. Hence, $\phi^*, \psi^* = n\pi$ for $n \in \mathbb{Z}$

$$\Rightarrow \underline{A}(\phi^*, \psi^*) = a \begin{pmatrix} \pm 1 \mp 2 & \pm 1 \mp 1 \\ \pm 1 \mp 1 & \mp 2 \pm 1 \end{pmatrix}$$

$$\text{e.g., } n=0 \text{ for } \phi^*, \psi^*, \Rightarrow \underline{A} = a \begin{pmatrix} -1-2 & 1-1 \\ 1-1 & -2-1 \end{pmatrix} = a \begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}$$

$$\Rightarrow \lambda_{1,2} = -3a \text{ for } \vec{e}_1, \vec{e}_2 \text{ being unit basis vectors } \hat{x} \text{ and } \hat{y}. \Rightarrow \text{stable} \checkmark$$

For the other solution $\phi^* = \psi^* = 2\pi/3$, we find

$$\underline{A}(\phi^*, \psi^*) = a \begin{pmatrix} 1/2 + 1 & 1/2 - 1/2 \\ 1/2 - 1/2 & 1 + 1/2 \end{pmatrix} = a \begin{pmatrix} 3/2 & 0 \\ 0 & 3/2 \end{pmatrix}$$

$$\Rightarrow \lambda_{1,2} = +3a/2 \text{ for } (\vec{e}_1, \vec{e}_2) \in \left(\hat{x}, \hat{y}\right) \Rightarrow \text{Unstable (inconsistent)}$$

Hence the stable, experimentally observed pairwise equilibrium solution is not stable according to our choice of H.

P1] c. [$\dot{\theta}_1 = \omega - H(\phi) - H(\pi + \phi)$
 $\dot{\theta}_2 = \omega + H(\phi) - H(\pi)$
 $\dot{\theta}_3 = \omega + H(\phi + \pi) + H(\pi)$

$$\dot{\phi} = \dot{\theta}_1 - \dot{\theta}_2 = H(\pi) - H(\pi + \phi) - 2H(\phi)$$

$$\dot{\pi} = H(\phi) - H(\pi + \phi) - 2H(\pi)$$

Figures are uploaded (parameter values in figure title, where
 $a1 \Rightarrow a=1$,
 $bP5 \Rightarrow b=0.5$, etc.)
 $[\phi: x\text{-axis}, \pi: y\text{-axis}]$

d. we add a small, even component
 via

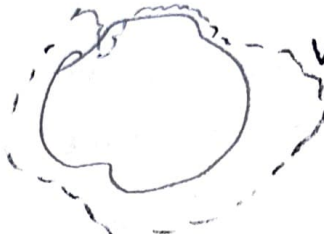
$$H(\theta) = a \sin \theta + b \sin 2\theta + \epsilon \cos \theta$$

where $\epsilon = 10^{-1}$

Figure is uploaded.

The even component, compared to ^{same} parameters without it, only acts to slightly deform trajectories away from the attractive manifolds / directions in phase space, but only slightly $\Rightarrow$ qualitatively the same

P2] a. an infinitesimal volume contracts according to the divergence of the vector field ^(or expands) to which it is subjected in phase space. For the Lorenz system $\vec{F}$, we have (consider a control volume V)

$V(t)$:  $V(t+\Delta t) \Rightarrow \dot{V} = \int_V \nabla \cdot \vec{F} dV$
 $\Rightarrow V(t) = V_0 e^{\nabla \cdot \vec{F} t}$

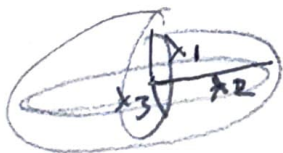
Lorenz system

$$\begin{cases} \dot{x} = \sigma(y-x) \\ \dot{y} = \rho x - y - xz \\ \dot{z} = xy - bz \end{cases} \Rightarrow \nabla \cdot \vec{F} = -\sigma - 1 - b$$

$$\Rightarrow V(t) = V_0 e^{(-\sigma - 1 - b)t}$$

Since $\sigma, b (\rho)$ are positive, $V(t)$ is an exponentially decaying volume. This implies that the attractor is not of the same dimension as our control volume V , but a lesser dimension (i.e., $\text{Dim}_{\text{attractor}} < 3$)

b. For the attracting set to be of dimension zero, we would need the volume to contract in all directions (eigenvectors). Our divergence calculation considers the net volume, but we need eigenvalues (Liapunov exponents) for each axis of the ellipsoidal control volume;

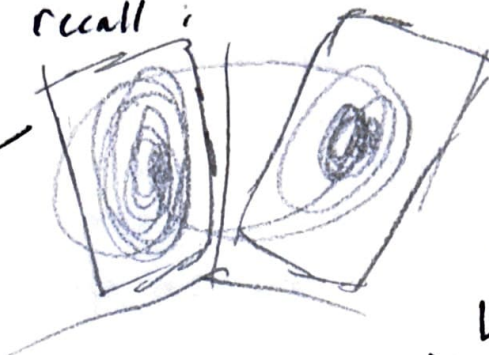


and all exponents λ_i would need to be negative. This occurs for attractive fixed points,

which the Lorenz system does not admit for typical parameters. But, perhaps it could for some (ρ, σ, B)

P2) c. We recall:
(2.3)

spirals appear
to be on
tilted "planes"



a curve erratically
bouncing between these
two spirals. We remember

The map (maxima in $\mathbb{Z}$ dir.)
Lorenz used worked because
it was roughly perpendicular
to the attractor. Since the

Spirals appear to reside on tilted planes (but we know
they actually have some thickness, due to the arguments of
Lorenz), We expect the attractor to be approximately
of dimension 2, but slightly larger, say

$$\dim \approx 2.1$$

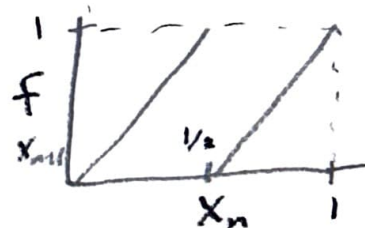
P3) a. Aperiodic, long-term behavior that exhibits
(P3.1) highly sensitive dependence on initial conditions
and never "settles down" (onto, e.g., fixed
points or limit cycles)

(P3.2) b. The Lyapunov exponent λ is defined (using chain
rule) as

$$\lambda_L = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right)$$

$$f(x_n) = x_{n+1} = 2x_n \text{ Mod } 1$$

graph of F



$\Rightarrow$ 2 consecutive linear curves in sequence

$\Rightarrow$ constant slope $\Rightarrow f'(x_n) = 2 \forall x_n$

$$\Rightarrow \lambda_L = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln |2| \right) = \lim_{n \rightarrow \infty} \left(\frac{1}{n} [n \ln |2|] \right) = \ln |2|$$

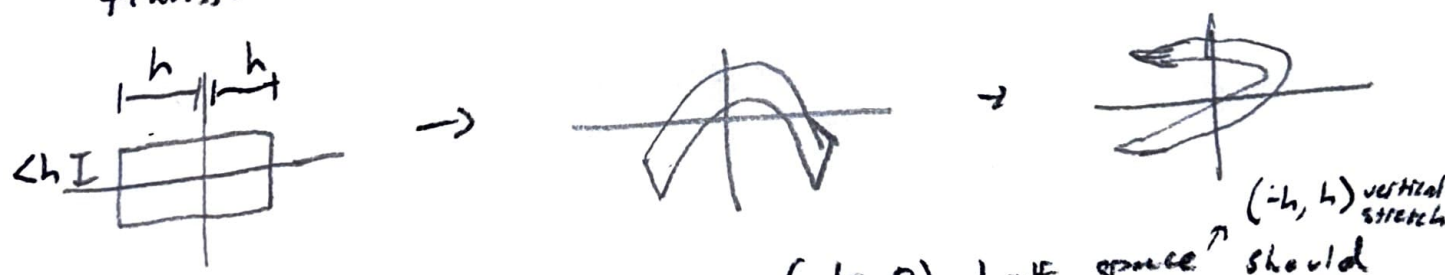
$$\Rightarrow \boxed{\lambda_L = \ln 2}$$

$$\lambda_c = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln |2| \right) = \ln |2| \quad \text{For}$$

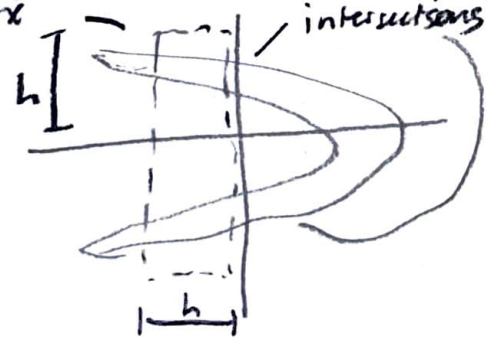
this Map as well.

(P3.4) d. Only the Map from part (b) is chaotic. The Modulo operator is the key; it restricts the Map image to the unit interval, so orbits stay bouncing around it forever. In contrast, without the modulo factor, all orbits tend toward infinity as the $n \rightarrow \infty$: this is an attracting fixed point, of sorts, contradicting the notion of never settling down/approaching a fixed point.

P4) a. The Henon map exhibits the fold and stretch transformations:



Hence, a box in the $(-h, 0)$ half spaceⁿ should be 'pierced' by both ends of the stretched and tilted initial surface. That is, our box intersections: 2 per iteration



$\Rightarrow$ after n iterations, 2^n intersections

b. $H \equiv \begin{cases} x_{n+1} = by_n + 1 - ax_n^2 \\ y_{n+1} = x_n \end{cases} \quad x_n = y_{n+1}$

$$\Rightarrow x_{n+1} = by_n + 1 - ay_{n+1}^2$$

$$\Rightarrow by_n + 1 = x_{n+1} + ay_{n+1}^2$$

$$\Rightarrow H^{-1} = \begin{cases} y_n = \frac{x_{n+1} + ay_{n+1}^2 - 1}{b} \\ x_n = y_{n+1} \end{cases}$$

invertible $\forall b \neq 0$

c. I don't have time to compute the backward- n iterations; so I expect/assume that doing so would produce 'opposing' folding and stretching, so that they may intersect $(2 \cdot 2)^n$ times, i.e., $4^n = 4^n$ intersections